CampusEvent

INITIAL PRODUCT BACKLOG & QA TEST PLANNING

Project: University Event Management and QR Ticketing Platform

1. Product Backlog Overview

CampusEvent is a web-based platform designed to centralize university event management, student registration, QR-code ticket generation, and attendance verification. The backlog identifies prioritized epics, features, user stories, acceptance criteria, effort estimates, dependencies, and responsible roles.

- Epic 1: User and Account Management
- Epic 2: Event Management
- Epic 3: Event Registration and Digital Ticketing
- Epic 4: Ticket Verification and Attendance Tracking

Priority: High = essential MVP functionality; Medium = important supporting functionality; Low = optional future improvements. Story points (SP) are preliminary relative complexity estimates.

2. Prioritized Product Backlog

ID	Epic	Feature	Priority	SP	Proposed owner
US-01	User Management	User Registration	High	5	QA & Test Automation Engineer
US-02	User Management	User Login	High	3	QA & Test Automation Engineer
US-03	Event Management	Event Creation	High	5	Implementation Engineer
US-04	Event Management	Event Browsing	High	3	Product Owner
US-05	Registration	Event Registration	High	5	Implementation Engineer
US-06	Ticketing	QR Ticket Generation	High	8	Software Architect
US-07	Ticket Verification	QR Ticket Verification	High	8	Software Architect
US-08	Attendance	Attendance Tracking	High	5	DevOps Engineer+ Implementation Engineer
US-09	Event Management	Event Search	Medium	3	Product Owner
US-10	Event Management	Event Editing	Medium	5	Implementation Engineer

Total preliminary estimate: 50 story points. Estimates must be reviewed collectively by the implementation team, Software Architect, and QA Engineer, then recorded by the Product Owner.

3. User Stories and Acceptance Criteria

Epic 1 – User and Account Management

US-01: User Registration

As a student, I want to create an account so that I can access CampusEvent.

- AC-01.1: Users can register with valid information.
- AC-01.2: Required fields cannot be empty.
- AC-01.3: Duplicate accounts are rejected.
- AC-01.4: Successful registration displays a confirmation message.

US-02: User Login

As a registered user, I want to log in securely so that I can access my account.

- AC-02.1: Valid credentials allow successful login.
- AC-02.2: Invalid credentials are rejected.
- AC-02.3: Unauthenticated users cannot access protected features.

Epic 2 – Event Management

US-03: Event Creation

As a club representative, I want to create events so that students can participate in campus activities.

- AC-03.1: Authorized club representatives can create events.
- AC-03.2: Event title, date, time, and location are required.
- AC-03.3: Incomplete event forms are rejected.
- AC-03.4: Unauthorized users cannot create events.

US-04: Event Browsing

As a student, I want to browse events so that I can discover campus activities.

- AC-04.1: Available events are displayed.
- AC-04.2: Event details include title, date, time, and location.
- AC-04.3: Students can view detailed event information.

US-09: Event Search

As a student, I want to search for events so that I can find relevant activities.

- AC-09.1: Users can search by event keywords.
- AC-09.2: Matching events are displayed.
- AC-09.3: A message appears when no matching events are found.

US-10: Event Editing

As a club representative, I want to update my events so that event information remains accurate.

- AC-10.1: Authorized club representatives can edit their events.
- AC-10.2: Changes are saved successfully.
- AC-10.3: Unauthorized users cannot modify events.

Epic 3 – Event Registration and Digital Ticketing

US-05: Event Registration

As a student, I want to register for events so that I can participate in university activities.

- AC-05.1: Logged-in students can register for available events.
- AC-05.2: Duplicate registrations are prevented.
- AC-05.3: Successful registration displays confirmation.
- AC-05.4: Failed registrations display appropriate errors.

US-06: QR Ticket Generation

As a registered student, I want to receive a QR-code ticket so that I can enter an event.

- AC-06.1: A QR-code ticket is generated after successful registration.
- AC-06.2: Each ticket is associated with the correct registration and event.
- AC-06.3: Students can access their digital tickets.

Epic 4 – Ticket Verification and Attendance Tracking

US-07: QR Ticket Verification

As an event organizer, I want to verify QR tickets so that only valid registrations are accepted.

- AC-07.1: Authorized organizers can verify tickets.
- AC-07.2: Valid tickets are accepted.
- AC-07.3: Invalid tickets are rejected.
- AC-07.4: Previously used tickets cannot be accepted again.

US-08: Attendance Tracking

As an event organizer, I want to track attendance so that participation records are maintained.

- AC-08.1: Successful ticket verification updates attendance status.
- AC-08.2: Attendance records are linked to the correct student and event.
- AC-08.3: Authorized organizers can view attendance records.

4. Dependencies

Story	Depends on
US-01	None
US-02	US-01
US-03	US-02
US-04	US-03
US-05	US-02, US-04
US-06	US-05
US-07	US-06
US-08	US-07
US-09	US-04
US-10	US-03

5. Initial QA Test Strategy

5.1 Testing Objective

Verify that CampusEvent functions correctly, securely, and consistently with approved acceptance criteria. Testing will focus on authentication, event management, registration, QR-code ticketing, and attendance tracking.

5.2 Testing Types

Unit Testing

- User input validation
- Duplicate registration checks
- QR ticket validation
- Attendance status updates

Integration Testing

- User authentication and database operations
- Event registration and ticket generation
- QR verification and attendance records

End-to-End Testing

- Student creates an account and logs in
- Student browses and registers for an event
- System generates a QR ticket
- Organizer verifies the ticket and attendance is recorded

5.3 Initial Test Scenarios

Test ID	Story	Test scenario	Expected result
TC-01	US-01	Register with valid information	Account created
TC-02	US-01	Register with duplicate information	Registration rejected
TC-03	US-02	Login with valid credentials	Login successful
TC-04	US-02	Login with invalid credentials	Login rejected
TC-05	US-03	Create event as authorized representative	Event created
TC-06	US-03	Create event with missing required fields	Validation error
TC-07	US-04	Browse available events	Events displayed
TC-08	US-05	Register for available event	Registration successful
TC-09	US-05	Register twice for same event	Duplicate rejected
TC-10	US-06	Generate QR after registration	QR ticket generated
TC-11	US-07	Verify valid QR ticket	Ticket accepted
TC-12	US-07	Verify invalid QR ticket	Ticket rejected
TC-13	US-07	Reuse already verified ticket	Ticket rejected
TC-14	US-08	Verify attendance after scanning	Attendance updated
TC-15	US-09	Search for existing event	Matching events displayed
TC-16	US-10	Edit event without authorization	Access denied

5.4 Defect Tracking

Defects will be recorded in GitHub Issues with defect ID/title, related story, reproduction steps, expected and actual results, severity, and status.

5.5 Test Coverage

The project will target at least 50% unit-test coverage during implementation, measured with tools appropriate to the selected technology stack.

5.6 Test Execution Status

All scenarios are Not Executed. Actual results, pass/fail statuses, and measured coverage will be recorded after implementation and test execution.

6. Definition of Ready

- User role and expected value are clear.
- Acceptance criteria are specific and testable.
- Dependencies and API impact are understood.
- Team has reviewed the effort estimate.
- Story can be completed within one sprint.
- An implementation owner is assigned.

7. Definition of Done

- Required functionality is implemented.
- Acceptance criteria are satisfied.
- Relevant tests have passed.
- Critical defects are resolved.
- Changes are peer-reviewed through a pull request.
- Documentation is updated where necessary.

8. Roles and Responsibilities

Product Owner & Requirements Engineer: Reviews backlog priorities and validates stories and acceptance criteria.

QA & Test Automation Engineer: Reviews testability, prepares scenarios and strategy, tracks defects and coverage.

Software Architect & Technical Lead: Reviews dependencies, API consistency, and feasibility.

Implementation & Integration Engineer: Implements assigned functionality and supports testing.

DevOps & CI/CD Engineer: Maintains repository workflow and automated test infrastructure.

Scrum Master: Coordinates sprint planning and tracks progress.